(ii) $R_2 = \{(0,10),(1,8),(2,6),(2,4),(3,4),(4,2)\}$

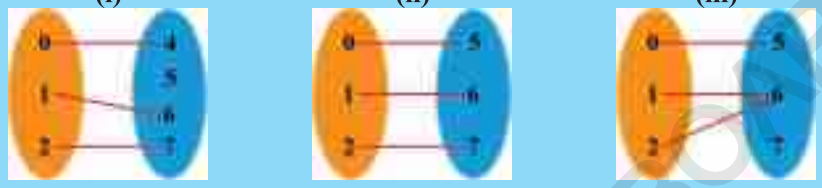
(iii) $R_3 = \{(0,2),(1,4),(2,6),(3,8),(4,8)\}$

(iv) $R_4 = \{(0,2),(1,2),(2,2),(3,2),(4,8)\}$

(v) $R_5 = \{(0,2),(1,4),(2,6),(3,8),(4,10)\}$

11. Which of the following is one-one function or one-one correspondence or neither of them

(i) **(ii)** **(iii)**



12. If $P = \{a,b,c\}$, $Q = \{x,y,z\}$ and $R = \{p,q,r,s\}$ then find:

(i) a function f from P into Q.
(ii) a function g from R onto P.
(iii) a function h from P to R which is injective.
(iv) a function k from Q to P which is bijective.

Review Exercise 17

1. Choose the correct option:

(i) Which one is subset of N

(a) Z (b) Q (c) E (d) P

(ii) Tabular form of $A = \{x \mid x \in Z \wedge x^2 = 25\}$ is _____.

(a) $\{1,5\}$ (b) $\{-5,5\}$

(c) $\{1,5,25\}$ (d) $\{-5\}$

(iii) For two sets A and B, if $O(A) = O(B)$ then

(a) $A \subseteq B$ (b) $B = A$

(c) $A \sim B$ (d) $A \subset B$

(iv) $\{x \mid x \in N \wedge x < 1\}$ is _____

(a) Singleton (b) Infinite set

(c) Empty set (d) Super set

(v) If $A = \{1,2,3,4\}$ and $B = \{2,4,6\}$ then $A \Delta B =$ _____.

(a) $\{1,2\}$ (b) $\{6\}$

(c) $\{1,3\}$ (d) $\{1,3,6\}$

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(vi) Which operation is not commutative

(a) Symmetric difference (b) Union

(c) Difference (d) Intersection

(vii) $\{x \mid x \in P \vee x \in Q\} =$ _____.

(a) $P \cap Q$ (b) $P \cup Q$

(c) $P - Q$ (d) $P \Delta Q$

(viii) $\{x \mid x \in U \wedge x \notin A \cap B\} =$ _____.

(a) $A \cap B$ (b) $(A \cup B)'$

(c) $(A \cap B)'$ (d) $A' \cap B'$

(ix) For any two non-empty sets A and B, if $A \cup B = U$ and $A \cap B = \varnothing$ then A and B are _____.

(a) Cells (b) Overlapping sets

(c) Equal sets (d) None

(x) $\left((A')'\right)' =$ _____.

(a) $(A')'$ (b) A'

(c) A (d) B

(xi) If $A \supset B$ then $A \cup B =$ _____.

(a) A (b) B

(c) $\varnothing$ (d) U

(xii) In Venn diagram, _____ is used to represent universal set.

(a) rectangle (b) circle

(c) oval (d) all of these

(xiii) If $(x, 6) = (2, y - 6)$ then $x + y =$ _____.

(a) 8 (b) 10 (c) 12 (d) 14

(xiv) If $O(A \times B) = 100$ and $O(B) = 5$ then $O(A) =$ _____.

(a) 10 (b) 15 (c) 20 (d) 25

(xv) A function is called surjective if range _____ co-domain

(a) $\supset$ (b) $\subset$

(c) $=$ (d) $\neq$

(xvi) A function is called into if range _____ co-domain

(a) $\supset$ (b) $\subset$

(c) $=$ (d) none of these

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(xvii) Given that $A = \{0,1,2\}$ and $B = \{3,4\}$ then decide which of the followings is a function $f: A \to B$?

(a) $\{(0,3),(1,4)\}$ (b) $\{(0,3),(1,4),(2,3),(2,4)\}$

(c) $\{(0,3),(1,4),(2,4)\}$ (d) none of these

(xviii) Which one is always bijective

(a) Onto function (b) One-one function

(c) Into function (d) One-one correspondence

(xix) If $f: \mathbb{N} \to \mathbb{R}$ then which one is not possible

(a) $f(5) = 6$ (b) $f(7) = 8$

(c) $f(-2) = 6$ (d) $f(9) = 0$

(xx) Which one of the followings is a commutative set operation?

(a) Difference (b) Union

(c) Cartesian product (d) None

2. If $A = \{1,2,5\}$, $B = \{2,4,6\}$, $U = \{1,2,3,4,5,6\}$, then find the following

(a) $A \cup B$ (b) $A \cap B$ (c) $A \Delta B$

(d) $A - B$ (e) A' (f) B'

3. Verify De Morgan's laws for the sets of question 2.

4. Verify commutative properties of union and intersection for the sets.

$A = \{a,b,d\}$ and $B = \{a,d,e\}$

5. For sets $P = \{1,3,5\}$ and $Q = \{1,3,7\}$ represent the following set by using Venn diagram.

(a) $P \cup Q$ (b) $P \cap Q$ (c) $P \Delta Q$ (d) $P - Q$

6. If $A = \{a,c\}$ and $B = \{b,d\}$ then find

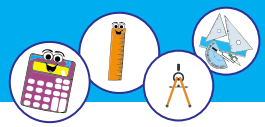
(a) Two relations in A (b) Three relations in B

(c) Three relations in $A \times B$ (d) Two relations in $A \cup B$

7. If $A = \{1,2,5\}$ and $B = \{6,8\}$ then find function from A to B, which is

(a) Onto function (b) Into function

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Example 4:

The expenses of a hostel are partly constant and partly vary according to the number of students. When the number of students are 140 the expenditure is Rs 19300 and for 200 students the expenditure is Rs 26500. Find the expenses when number of students are 300.

Solution: Let the constant expenses be denoted as C, Q denotes partly expense, x is the number of students and P denotes the total expense.

Since the expenses varies directly to number of students.

i.e., $Q \propto x$

$Q = nx$ where n is constant

$P = C + Q$

$P = C + nx$ where n is constant

According to given condition

Hence $19300 = C + 140n$ ...(i)

and $26500 = C + 200n$ ...(ii)

also $P = C + 300n$ ...(iii)

From (i) and (ii), we get $n = 120$ and $C = 2500$

Now, equation (iii) becomes

$$P = 2500 + 300(120) = 38500$$

The total expense for 300 students is 38500.

Exercise 18.7

1. The current in a wire is directly proportional to the potential difference and inversely proportional to the resistance R. If I=6 amperes, when V=220 volts and R=5 ohms. Find I when V=180 volts and R=8 ohms
2. The intensity of light measured in foot candles varies inversly with the square of the distance from the light source. Suppose the intensity of a high bulb is 0.08 foot candles at a distance of 3 meters .find the intensity level at 8 meters.
3. The strength S of a rectangular beam can varies directly as the breadth b and the square of the depth d. if a bean 9 cm wide a 12 cm deep with support 1200 pound. What weight a bean of 12 cm wide and 9cm deep will support.
4. Labor cots C varies jointly as the number of worker n and the average number of days d, if the cost of 100 workers for 15 days is 9000, then find the labour cost of 300 workers for 20 days.
5. The sales tax on the purchase of a new car varies directly as the price of a car. If a new car is purchased in Rs 2000000, then the sale tax is Rs 40000. How much sale tax is charged if the new car is priced at Rs 2800000.

Review Exercise 18

1. Multiple Choice Question

Select the correct option in the following

(i) In a proportion $p : q :: r : s$, p is called

(a) third proportional (b) mean (c) fourth proportional (d) first proportional

(ii) In a ratio $l : m$, l is called

(a) consequent (b) antecedent (c) relation (d) none of these

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(iii) In a ratio $u:v$, v is called

(a) antecedent (b) consequent (c) relation (d) none of these

(iv) If a,b,c are in continued proportion then b is called _____ proportion between a & c

(a) 1st (b) mean (c) 3rd (d) None of these

(v) The mean proportional between a^2 and b^2 is _____

(a) $\sqrt{ab}$ (b) ab (c) $\frac{a}{b}$ (d) $-ab$

(vi) If $x+5:x+7=5:7$ then x is equal to _____

(a) 2 (b) -1 (c) 0 (d) 1

(vii) If 1,9, x and 45 are in proportion, then $x=$

(a) 27 (b) $\frac{1}{5}$ (c) 5 (d) 405

(viii) If $p:q=r:s$ then $p:r=q:s$ this property is called

(a) compodendo (b) invertendo (c) dividendo (d) alternando

(ix) If $x:y=z:w$ then according to componendo

(a) $\frac{x}{x+y}=\frac{z}{z+w}$ (b) $\frac{x}{x-y}=\frac{z}{z-w}$

(c) $\frac{x-y}{x+y}=\frac{z-w}{z+w}$ (d) $\frac{x-y}{y}=\frac{z-w}{w}$

(x) If $a:b=c:d$ then according to alternando property

(a) $\frac{a}{b}=\frac{c}{d}$ (b) $\frac{a+b}{b}=\frac{c+d}{c}$ (c) $\frac{b}{a}=\frac{d}{c}$ (d) $\frac{a}{c}=\frac{b}{d}$

(xi) The fourth proportional to 3,5,12 is

(a) 20 (b) 15 (c) 60 (d) 36

(xii) If $2x$, $3y$ and $6z$ are in continued proportion then

(a) $y^2=12xy$ (b) $9y^2=xz$ (c) $9y^2=12xz$ (d) $3y^2=4xz$

(xiii) If $\frac{x}{y}=\frac{w}{z}$ then according to dividendo property is

(a) $\frac{x-y}{y}=\frac{w-z}{z}$ (b) $\frac{x+y}{y}=\frac{w+z}{z}$

(c) $\frac{x}{x+y}=\frac{w}{w+z}$ (d) None of these

(xiv) Force and acceleration are in

(a) direct proportion (b) joint proportion

(c) inverse proportion (d) None of these

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(xv) If $a:4::15:5$ then a =_____

(a) 20 (b) 15 (c) 12 (d) 10

2. Find the ratios of the

(i) 100m and 500cm (ii) 50kg and 300g

3. Find the value of x in the following

(i) $5:x-3=x+11:3$ (ii) $9:x-10=x+13:12$

4. If y varies directly as x and $y = 25$ when $x = 75$ then find y when $x = 144$.

5. If y varies inversely as x and $y = 100$ when $x = \frac{1}{2}$ then find y when $x = 4$.

6. If $x : y = z : w$ then prove that $\frac{7x+5y}{7x-5y}=\frac{7z+5w}{7z+5w}$.

7. Solve $\frac{(x-3)(x-5)}{(x-7)(x-2)}=\frac{(x-6)(x-2)}{(x-1)(x-8)}$ by componendo – dividendo theorem.

8. If x varies directly as y and inversely as z. If $x = 30$ when $y = 15$ and $z = 2$. Find x if $y = 30$ when $z = 12$.

9. The current in a circuit varies inversely with its resistance measured in ohms. When the current in a circuit is 40 ampere, the resistance in 10ohms. Find the current if resistance is 12 ohm.

Summary

- The comparison between two quantities of the same kind is called ratio
- If two ratio $a:b$ and $c:d$ are equal then we can write.
  $a:b::c:d$
- Equality of two ratios is called proportion.
- If two quantities are related in such a way that increase (decrease) in one quantity causes increase (decrease) respectively in the other quantity is called direct variation.
- If two quantities are related in such a way that when one quantity increase, then the other decreases and vice versa is called the quantities are in direct variation.
- Theorems on proportion

  (i) Theorems invertando If $a:b=c:d$ then $b:a=d:c$

  (ii) Theorems of alternendo If $a:b=c:d$ then $a:c=b:d$

  (iii) Theorems of componendo

  If $a:b=c:d$ then (*a*) $a+b:b=c+d:d$ (*b*) $a:a+b=c:c+d$

  (iv) Theorem of dividend

  If $a:b=c:d$ then $a-b:b=c-d:d$ and $a:a-b=c:c-d$

  (v) Theorem componendo and dividendo

  If $a:b=c:d$ then

  $a+b:a-b=c-d:c-d$
- If one variable varies directly or inversely with two or more than two variables then it is called joint variation

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iii) $C=\begin{bmatrix}2 & 1 & 1\\ 3 & 2 & 1\\ 2 & 1 & 2\end{bmatrix}$ iv) $D=\begin{bmatrix}1 & 0 & 1\\ -4 & 1 & -1\\ 6 & -2 & 1\end{bmatrix}$ v) $E=\begin{bmatrix}1 & -1 & 3\\ 2 & 1 & 2\\ -2 & -2 & 1\end{bmatrix}$.

8. Find the solution by matrix inversion method and cramer's rule:

i) $2x+3y=14$, $-4x+y=28$ ii) $2x-4y=-12$, $2y+3x=0$

Review Exercise 19

1. Tick the correct option

i. If m denotes the number of rows and n denotes the number of columns such that $m=n$, then matrix is called _____ matrix.

(a) Rectangular (b) Equal (c) Square (d) Null

ii. If $A=\begin{bmatrix}i & 0\\ 0 & -i\end{bmatrix}$ then A^2 is

(a) I_2 (b) I_3 (c) $-I_2$ (d) O

iii. If A is any square matrix such that $A^t=-A$, then A is said to be:

(a) Diagonal matrix (b) Scalar matrix
(c) Symmetric matrix (d) Skew Symmetric matrix

iv. If A , B and C are matrices of same order then $(ABC)^t=$

(a) $A^t.B^t.C^t$ (b) $C^tB^tA^t$
(c) $C^tA^tB^t$ (d) $(B^tA^t)C^t$

v. If $A=\begin{bmatrix}i & 0\\ 0 & -i\end{bmatrix}$, $B=\begin{bmatrix}0 & i^2\\ -i^2 & 0\end{bmatrix}$ and $C=\begin{bmatrix}0 & i\\ i & 0\end{bmatrix}$ then

(a) $A^2=-I_2$ (b) $B^2=-I_2$
(c) $C^2=-I_2$ (d) All of them

vi. For two matrices A and B if $A=\begin{bmatrix}\sin\theta & -\cos\theta\\ \cos\theta & \sin\theta\end{bmatrix}$ and $B=\begin{bmatrix}\sin\theta & \cos\theta\\ -\cos\theta & \sin\theta\end{bmatrix}$ then AB =

(a) $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ (b) $\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}$
(c) $\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$ (d) $\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$

vii. If $2\begin{bmatrix}x\\ 3\\ 4\end{bmatrix}+3\begin{bmatrix}-1\\ y\\ 0\end{bmatrix}=4\begin{bmatrix}0\\ 4\\ z\end{bmatrix}$ then the values of x, y and z respectively are_____

(a) $2,\frac{3}{2},\frac{10}{3}$ (b) $\frac{3}{2},2,\frac{10}{3}$
(c) $\frac{3}{2},\frac{10}{3},2$ (d) 1,2,3

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viii. For matrix A, $\left(A^{-1}\right)^{-1} =$ ——

(a) A^{-2} (b) A (c) A^{-1} (d) A^2

ix. Find x if $\begin{vmatrix} 5 & 1 \\ 2 & x \end{vmatrix} = x + 4$

(a) $\frac{3}{2}$ (b) $\frac{2}{3}$ (c) 0 (d) None of them

x. If the matrix $\begin{bmatrix} \lambda & -3 & 4 \\ -3 & 0 & 1 \\ -1 & 3 & 2 \end{bmatrix}$ is invertible then $\lambda \neq$ ——

(a) -15 (b) -17 (c) -16 (d) None of these

2. Define the row and column of a matrix.

3. Find the order of the matrices:

i) $\begin{bmatrix} 1 & 5 & 8 \end{bmatrix}$ ii) $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 8 \end{bmatrix}$ iii) $\begin{bmatrix} 7 & 2 \\ 9 & 3 \\ 8 & 1 \end{bmatrix}$ iv) $\begin{bmatrix} a+b+c \end{bmatrix}$

4. Define the following matrices:

i) Square matrix ii) Rectangular matrix
iii) Diagonal matrix iv) Scalar matrix
v) Symmetric matrix vi) Skew Symmetric matrix

5. If $A = \begin{bmatrix} 1 & 8 & 9 \\ 2 & 1 & 0 \\ -2 & 1 & 7 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 5 & 8 \\ 1 & 3 & 4 \\ 7 & 2 & 4 \end{bmatrix}$ then find

i) $A + B$ ii) $A - B$ iii) AB iv) BA
v) $5A$ vi) $7B$ vii) $8A - 9B$

6. Evaluate $\begin{vmatrix} 1 & 8 & 9 \\ 2 & 0 & -1 \\ -7 & 8 & -10 \end{vmatrix}$

7. Define singular and non-singular matrices with examples.

8. If $A = \begin{bmatrix} 1 & 4 & 2 \\ 7 & 0 & 9 \\ 0 & 2 & -3 \end{bmatrix}$ then find A^{-1}

9. Solve $2x + 5y = 27$

$7x + y = 12$

by i) matrix inversion method ii) Cramer's rule

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Review Exercise 20

1. Multiple Choice Question MCQs.

Encircle the correct answer.

i. If p, q are the roots of $2x^2+5x+3=0$, then $p+q=$ ———

(a) $\frac{5}{3}$ (b) $\frac{3}{5}$ (c) $\frac{5}{2}$ (d) $-\frac{5}{2}$

ii. If $\frac{1}{\alpha}, \frac{1}{\beta}$ are the roots of the $ax^2+bx+c=0,\ a\neq 0$, then $\alpha+\beta=$ ———

(a) $-\frac{b}{a}$ (b) $\frac{b}{c}$ (c) $-\frac{b}{c}$ (d) $-\frac{c}{b}$.

iii. If the sum of the roots of $(p+1)x^2+(2p+3)x+(3p+4)=0$ is -1, then product of the root is

(a) 0 (b) 1 (c) 2 (d) 3

iv. The nature of the roots of $ax^2+bx+c=0,\ a\neq 0$ is determined by

(a) sum of the roots (b) product of the roots
(c) discriminant (c) none of these

v. If sum of the roots of a quadratic equation is $\frac{b}{a}$ and the product of the roots is $\frac{c}{a}$, then equation is

(a) $ax^2+bx+c=0$ (b) $ax^2+bx-c=0$
(c) $ax^2-bx+c=0$ (d) $ax^2-bx-c=0$

vi. The required equation whose roots are the reciprocal of the roots of $ax^2+bx+c=0$ is

(a) $ax^2+bx+c=0$ (b) $cx^2+bx+a=0$
(c) $cx^2+ax+b=0$ (d) $cx^2-bx-a=0$

vii. If α, β are the roots of $x^2-2x-15=0$, then the value of $\alpha^2+\beta^2=$ ———

(a) 34 (b) -34 (c) 26 (d) -26

viii. If $\Delta=b^2-4ac$ of a quadratic equation with rational coefficients is perfect square, then roots are

(a) real and equal (b) real, rational and un equal
(c) real, irrational and unequal (d) imaginary or complex and unequal

ix. If one root of quadratic equation with rational coefficients is $2+\sqrt{3}$, then other root will be

(a) 2 (b) $-2+\sqrt{3}$ (c) $2-\sqrt{3}$ (d) $-2-\sqrt{3}$

x. The quadratic equation whose roots are complex cube roots of umity is

(a) $x^2-x-1=0$ (b) $x^2-x+1=0$
(c) $x^2+x+1=0$ (d) $x^2+x-1=0$

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Exercise 21.4

Resolve the following into partial fractions

1. $\dfrac{x^2}{\left(x^2+1\right)^2\left(1-x\right)}$
2. $\dfrac{x^2+x+2}{x^2\left(x^2+3\right)^2}$
3. $\dfrac{x^2}{\left(1+x\right)\left(1+x^2\right)^2}$
4. $\dfrac{7}{\left(x+1\right)\left(2+x^2\right)^2}$
5. $\dfrac{49}{\left(x-2\right)\left(3+x^2\right)^2}$

Review Exercise 21

1. Tick the correct option

i. An improper fraction can be reduced into proper fraction by

(a) addition (b) multiplication
(c) subtraction (d) division

ii. Partial fractions of $\dfrac{x}{(x-a)(x-b)(x-c)}$ can have a form

(a) $\dfrac{A}{x+a}+\dfrac{B}{x+b}+\dfrac{C}{x+c}$ (b) $\dfrac{A}{x-a}+\dfrac{B}{x-b}+\dfrac{C}{x-c}$
(c) $\dfrac{A}{x+a}+\dfrac{B}{x-b}+\dfrac{C}{x+c}$ (d) None of these

iii. Find the partial fractions of $\dfrac{x-3}{x^3+3x}$ are _____.

(a) $\dfrac{-1}{x}-\dfrac{x-1}{x^2+3}$ (b) $\dfrac{1}{x}+\dfrac{x+1}{x^2+3}$
(c) $\dfrac{1}{x}-\dfrac{x+1}{x^2+3}$ (d) $\dfrac{-1}{x}+\dfrac{x+1}{x^2+3}$

iv. $\dfrac{x^3+1}{(x-1)(x+2)}$ is

(a) Proper fraction (b) An improper fraction
(c) An identity (d) A constant term

v. The fraction $\dfrac{2x+5}{x^2+5x+6}$ is known as:

(a) Proper (b) Improper
(c) Both proper and improper (d) None of these

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EXERCISE 22.5

1. Find range, variance, mean deviation and standard deviation of number of absentees in a class for last seven days: 3, 5, 3, 2, 4, 1, 8.

2. Find range, mean deviation, variance and standard deviations of scores of two batsmen in 6 innings. Who is more consistent player?

Batsman-A	12	15	0	185	7	19
Batsman-B	47	12	76	48	4	51

3. Compare variation using range and standard deviation of income and expenditure of six families. Also interpret the results.

Income (in thousand Rs.)	10	20	30	40	50	60
Expenditure (in thousand Rs.)	7	21	23	34	36	53

4. Goals scored by teams A and B in a football season are given. Use relative standard and mean deviations to find which team performed more consistently?

Goals scored	0	1	2	3	4
Number of games A played	27	9	8	5	4
Number of games B played	17	9	6	5	3

5. Find range, variance, standard deviation and mean deviation in mass of 50 blocks of metal as distributed in the following data:

Mass of block(in Kg)	7.1-7.3	7.4-7.6	7.7-7.9	8.0-8.2	8.3-8.5	8.6-8.8	8.9-9.1
Number of blocks	3	5	9	14	11	6	2

Review Exercise 22

1. Tick the correct answer.

(i). Arithmetic mean is based on the use of _____ observations of data.
(a). Middle (b). Extreme (c). All (d). None

(ii). The ungrouped data must be ordered first to find _____ .
(a). A.M. (b). Mode (c). Median (d). Range

(iii). The number of classes in a continuous frequency table lies between 5 and ___.
(a). 10 (b). 15 (c). 20 (d). 25

(iv). Mode of grouped data is obtained graphically using _____.
(a). Histogram (b). Polygon (c). Ogive (d). Bar chart

(v). If data has outliers, then _____ is misleading.
(a). A.M. (b). Range (c). Median (d). Mode

(vi). The A.M. of {0, 90, k, 10, 100} is 40, then k = ____ .
(a). 0 (b). 90 (c). 10 (d). 100

(vii). Median and quartiles are _____ in nature.
(a). Mathematical (b). Positional (c). Logical (d). None of these

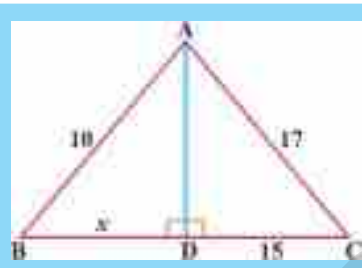
(viii). Upper quartile divides data in _____ ratio.
(a). 50%–50% (b). 25%–25% (c). 75%–25% (d). 40%–60%

(ix). If data contain a number equal to 0, then ____ cannot be computed.
(a). A.M. (b). G.M. (c). H.M. (d). Median

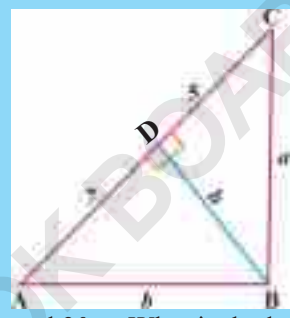
(x). If all numbers in data are equal, then:
(a). A.M.=G.M.=H.M. (b). Range = 0 (c). S.D. = 0 (d). All of these

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6. Find the value of x in the adjacent figure.



7. In the ΔABC, $m\angle B$ is a right angle and BD is a perpendicular on $\overline{AC}$ such that $m\overline{CD}=5$ units and $m\overline{AD}=7$ units as shown in the figure. Find the lengths of the unknowns a, h and b.



8. The sides of a rectangular swimming pool are 50m and 30m. What is the length between the opposite corners?

9. The length of each side of an equilateral triangle is 8 units. Find the length of any one altitude.

10. The sides of a triangle have lengths $x, x+4$ and 20. If the length of the longest side is 20. What values of x make the right triangle?

11. A manager goes 18m due east and then 24m due north. Find the distance of his current position from the starting point.

12. In the rectangle ABCD, $m\overline{BC}+m\overline{CD}=17\text{cm}$ and $m\overline{BD}+m\overline{AC}=26\text{cm}$ calculate the length and breadth of the rectangle.

REVIEW EXERCISE 23

1. Encircle the correct option

i. Diagonal of a rectangle measures 6.5cm. If its width is 2.5cm, its length is

(a) 9cm (b) 4cm

(c) 6cm (d) 3cm

ii. Which of the following are the sides of a right angled triangle?

(a) 3, 4, 5 (b) 2, 3, 4

(c) 5, 6, 7 (d) 4, 5, 6

iii. In a right angled triangle the greatest angle is

(a) 100° (b) 90°

(c) 80° (d) 110°

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iv. In a right angled triangle hypotenuse is opposite side to

(a) Acute angle (b) Right angle

(c) Obtuse angle (d) None

v. If a, b, c are sides of right angled triangle, with c is the larger side, then

(a) $c^2 = a^2 + b^2$ (b) $b^2 = c^2 + a^2$

(c) $a^2 = b^2 + c^2$ (d) $c^2 = a^2 - b^2$

vi. If 5cm and 12cm are two sides of a right angled triangle. Then hypotenuse is

(a) 16 (b) 15

(c) 14 (d) 13

vii. If hypotenuse of an isosceles right-angled triangle is $3\sqrt{2}$cm, then each of other side is of length

(a) 2cm (b) 5cm

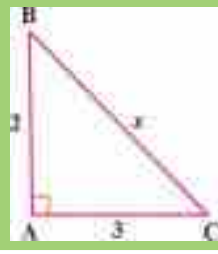
(c) 3cm (d) 1cm

2. Explain the Pythagoras theorem.

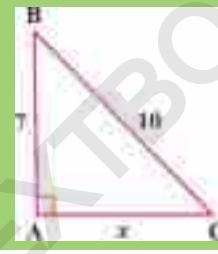
3. A ladder 25m long rests against a vertical wall. The foot of the ladder is 7m away from the base of the wall. Find the height of the wall given that ladder reaches at its top end.

4. Find the unknown values in each of the following figures

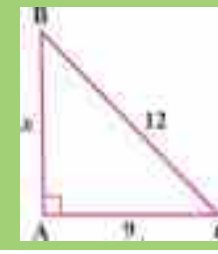
i)



ii)



iii)



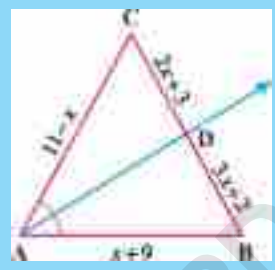
SUMMARY

- In a right angled triangle, the square of the length of hypotenuse is equal to the sum of the squares of the lengths of the other two sides.
- If the square of the one side of a triangle is equal to the sum of the square of the other two sides then the triangle is a right angled triangle.

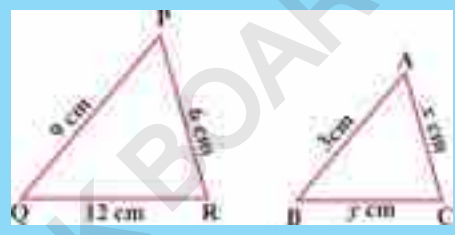
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EXERCISE 24.2

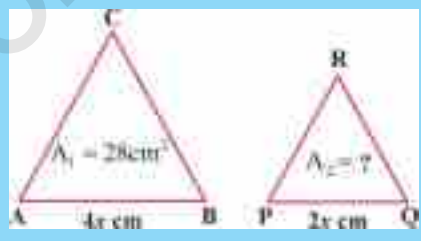
1. In the adjacent figure, $\overrightarrow{AD}$ is the bisector of $\angle A$ of ΔABC. Find the value of x if $m\overline{AB} = x+9, m\overline{AC} = 11-x, m\overline{CD} = 2x+3$ and $m\overline{BD} = 3x+2$. Also specify the type of triangle.



2. In the adjacent figure, ΔPQR and ΔABC are similar. Find the values of x and y if lengths of sides are indicated in the figure.



3. Let A_1 and A_2 be the areas of two similar triangles ΔABC and ΔPQR respectively as shown in the figure. Find A_2 if $A_1 = 28cm^2, m\overline{AB} = 4x$ cm and $m\overline{PQ} = 2x$ cm



4. Ratio of corresponding sides of two similar triangles is $2:x-5$ and the ratio of their areas is $1:9$. Find the value of x.

5. Prove that two right triangles have their sides proportional if an acute angle of the one is congruent to an acute angle of the other.

6. In a right triangle the perpendicular drawn from the right angle to the hypotenuse divides the triangle into two triangles. Prove that each of these triangles is similar to the original one.

REVIEW EXERCISE 24

1. Tick the correct option.

i. In a proportion, the product of means is equal to _____ of extremes.
(a) sum (b) difference (c) quotient (d) product

ii. _____ triangles are always similar.
(a) right (b) scalene (c) acute angled (d) equilateral

iii. The symbol of similarity of triangles is _____.
(a) = (b) ≅ (c) ~ (d) ↔

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iv. If a line parallel to base of a triangle and divides one side in 2:3 then it will divide other side in _____.

(a) 3:2 (b) 2:3 (c) 2:6 (d) 5:3

v. If a line intersects two sides of a triangle in same ratio then it is _____ to other side.

(a) parallel (b) non-parallel (c) coincident (d) all of these

vi. In ΔABC, the bisector of $\angle A$ divides $\overline{BC}$ in ratio _____if $m\overline{AB} = 6\text{cm}$ and $m\overline{AC} = 8\text{cm}$.

(a) 5:8 (b) 3:4 (c) 1:1 (d) 5:7

vii. The bisector of an angle of equilateral triangle divides the opposite side in

(a) 2:3 (b) 3:2 (c) 1:1 (d) 5:2

viii. The corresponding sides of two similar triangles are_____.

(a) equal (b) un- equal (c) proportional (d) None of these

ix. If the ratio of two corresponding sides of similar triangle is 5:7 then ratio of their areas is equal to _____.

(a) 5:7 (b) 7:5 (c) 25:7 (d) 25:49

x. If the ratio of areas of two similar triangles is 36:121 then the ratio of its corresponding sides will be _____.

(a) 6:10 (b) 6:11 (c) 11:6 (d) 10:6

xi. If the ratio of corresponding sides of similar triangles is 2:x and that of areas is 4:9 then $x =$ _____.

(a) 3 (b) -3 (c) both a and b (d) none of these

xii. Two equilateral triangles are also _____.

(a) congruent (b) similar (c) proportional (d) equivalent

SUMMARY

- Ratio is the comparison of two similar quantities.
- In ratio $a:b$, a is called antecedent and b is called consequent.
- Proportion is the equality of two ratios.
- In proportion, the product of means is equal to the product of extremes.
- Two triangles are similar if they are equiangular.
- If two triangles are similar then their corresponding sides are proportional.
- A line parallel to one side of a triangle and intersecting the other two sides, divides them proportionally.
- If a line segment intersects the two sides of a triangle in the same ratio then it is parallel to the third side.
- The internal bisector of an angle of a triangle divides the side opposite to it in the ratio of the sides containing the angle.

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Review Exercise 25

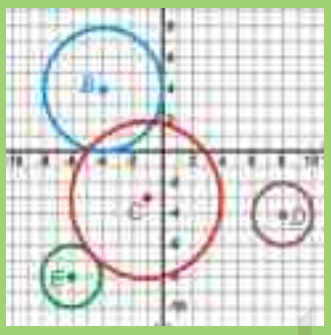
1. Tick the correct option.

i. All circles are always _____ .

(a) congruent (b) similar

(c) tangent (d) none of these

ii. In the following figure, circles centred at D and E are _____ .



(a) congruent (b) similar

(c) both (a) and (b) (d) none of these

iii. In the given figure, the length of the chord $\overline{AB}$ is _____ .



(a) 4cm (b) 6cm (c) 8cm (d) 15cm

iv. One and only one circle passes through three _____ points.

(a) collinear (b) non-collinear (c) disjoint (d) none of these

v. The hypothesis of the enunciation "If two chords of a circle are congruent, then they are equidistant from the centre." is:

(a) two chords of a circle are equidistant from the centre.

(b) two chords of a circle are congruent.

(c) a circle has two chords.

(d) the centre of circle is equidistant from chords.

vi. The conclusion of the enunciation "If two chords of a circle are congruent, then they are equidistant from the centre." is:

(a) two chords of the circle are equidistant from the centre.

(b) two chords of a circle are congruent.

(c) a circle has two chords.

(d) the centre of circle is equidistant from chords.

vii. The hypothesis of the enunciation "One and only one circle can pass through three non-collinear points" is:

(a) three points are non-collinear.

(b) one and only one circle passes through three points.

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(c) two circles pass through three points.
(d) three points are collinear.

viii. The conclusion of the enunciation "One and only one circle can pass through three non-collinear points" is:
(a) three points are non-collinear.
(b) one and only one circle passes through three points.
(c) two circles pass through three points.
(d) three points are collinear.

ix. A circle is an example of a _____ curve.
(a) simple and closed (b) simple and open
(c) non-simple and closed (d) non-simple and open

SUMMARY

- A circle is a simple closed curve.
- All points of a circle are equidistant from a fixed point. The fixed point is the centre and constant distance is radius of circle.
- A radial segment in a circle is a line segment from its centre to any point on it.
- A chord is a line segment joining any two points of a circle. A chord passing through the centre is known as diameter or central chord of the circle.
- For a circle of radius r, its diameter, circumference and circular area are equal to: $2r$, $2\pi r$ and πr^2, respectively.
- A portion of circumference of a circle is called an arc. These are further categorized as minor and major arcs.
- Sectors are portions of circle bounded by two radii and the intercepted arc. These can be further categorized as minor and major sectors.
- Segments are portions of circle bounded between a chord and the intercepted arc. These are further divided into major and minor sectors.
- All circles are similar to each other.
- Two circles are congruent if their radii are equal.
- Circles having same centre are concentric.
- Points lying on a circle are concyclic.
- One and only one circle can pass through three non-collinear points.
- A straight line, drawn from the centre of a circle to bisect a chord (which is not a diameter) is perpendicular to the chord.
- Perpendicular from the centre of a circle to a chord bisects it.
- If two chords of a circle are congruent then they will be equidistant from the centre.
- Two chords of a circle which are equidistant from the centre are congruent.

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6. If two chords in a circle of radius 2.5cm are 3.9cm apart, and length of a chord is 1.4cm, then find the length of the other chord.
7. If three circles touch in pairs externally and radii of two circles are 4cm and 5cm, then find the radius of the third circle if the triangle formed through their centres has perimeter of 35cm.

Review Exercise 26

1. Tick the correct answer.

i. If a point is outside the circle then from this point we can draw ____ tangent(s) to the circle.
(a) one (b) two (c) three (d) none

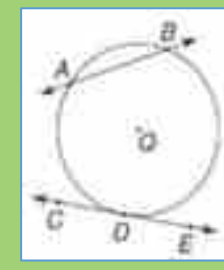
ii. Angle between the radial segment and tangent at its outer end point is ____ .
(a) 45^0 (b) 60^0 (c) 90^0 (d) 120^0

iii. In the adjacent figure, circles with centres E and C __ .
(a) touch internally (b) touch externally
(c) do not touch (d) are congruent



iv. In the adjacent figure, circles with centres B and C have _____ point(s) of contact.
(a) no (b) one
(c) two (d) none of these

v. In adjacent figure, circles with centres E and C have _____ point(s) of contact.
(a) no (b) one
(c) two (d) none of these



vi. In the adjacent figure, AB is _____ .
(a) tangent (b) secant
(c) chord (d) none of these

vii. In the adjacent figure, CDE is _____ .
(a) tangent (b) secant
(c) chord (d) none of these

viii. In the adjacent figure, the point of tangency is __ .
(a) A (b) B (c) C (d) D

ix. Tangents drawn at end points of a diameter of a circle are _____ to each other
(a) parallel (b) perpendicular (c) intersecting (d) both (b) and (c)

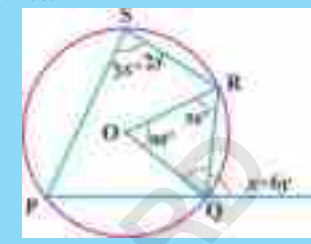
x. The maximum number of common tangents between two circles touching internally is ____ .
(a) 0 (b) 1 (c) 2 (d) 3

xi. The maximum number of common tangents between two circles touching externally is ____ .
(a) 0 (b) 1 (c) 2 (d) 3

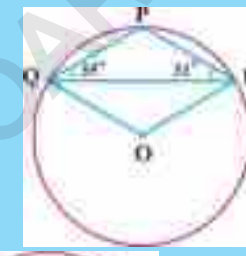
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EXERCISE 27.1

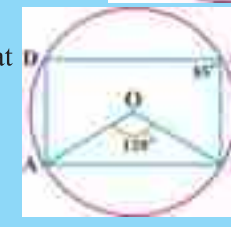
1. In a circle prove that the arcs between two parallel and equal chords or equal.
2. Prove that in equal circles, equal central angles have equal arcs.
3. In the following figure, O is the centre of the circle. Find the value of x and y.



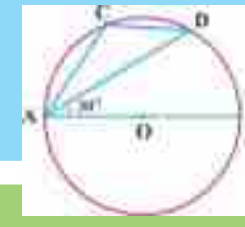
4. In the following figure $m\angle PQR = 35°$ and $m\angle PRQ = 31°$. Find $m\angle QPR$ and $m\angle OQR$.



5. In the figure O is the center of circle, given that $m\angle AOB = 120°$ and $m\angle BCD = 85°$. Find $m\angle OAD$



6. In the following figure, $\overline{AB}$ is diameter of the circle given that $m\angle DAB = 30°$ Find $m\angle ACD$



REVIEW EXERCISE 27

1. Multiple Choice Question:
Tick the correct option.

i. If a chord of a circle subtends, a central angle of $60°$. Then the chord and the radial segment are _____.
(a) Parallel (b) Perpendicular (c) Congruent (d) Incongruent

ii. An arc subtends a central angle of $45°$ then the corresponding chords will subtend a central angle of ?
(a) $15°$ (b) $30°$ (c) $45°$ (d) $60°$

iii. A pair of chords of a circle subtending two congruent central angle are.
(a) Perpemdicular (b) Non congruent (c) Congruent (d) None of these

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iv. The arcs opposite to congruent central angles of a circle are always.
(a) Parallel (b) Congruent (c) Perpendicular (d) None of these

v. A 6cm long chord subtends a central angle of 60°. The radial segment of this circle is
(a) 4cm (b) 6cm (c) 5cm (d) 8cm

vi. The chord length of a circle subtending a central angle of 180° is always
(a) equal to the radial segment
(b) less than radial segment
(c) double of radial segment
(d) half of the radial segment.

viii. The length of a chord and the radial segment of a circle are congruent, the central angle made by the chord will be:
(a) 60° (b) 75° (c) 90° (d) 45°

ix. Out of two congruent arcs of a circle, if one arc makes a central angle of 30° then the other arc will subtend the central angle:
(a) 60° (b) 90° (c) 75° (d) 30°

x. Diameter divides the circle into _____ parts.
(a) Two (b) Three (c) Four (d) All of these

SUMMARY

- The circles are congruent of their radii are equal.
- Equal chords of circle subtend equal angles at the centre.
- The straight line joining any two points of the circumference is called a chord of the circle.
- The portion of a circle bounded by an arc and a chord is known as the segment of a circle.
- The boundary traced by a moving point in a circle is called its circumference.
- An portion of the circumference of a circle is called an arc of the circle.
- A straight line, drawn from the centre of a circle bisecting a chord is perpendicular to the chord
- A perpendicular drawn from the centre of a circle on a chord, it bisect the chord.
- If two arcs of a circle(or of congruent circles) are congruent, then the corresponding chords are equal.
- If two chords of a circle (or of congruent circles) are equal, then their corresponding arcs (minor, major, semi circular) are congruent.
- Equal chords of a circle (or of congruent circles) subtend equal angles at the centre (at the corresponding centres).
- If the angle subtended by two chords of a circle (or congruent circles) at the centre (corresponding centres) are equal, the chords are equal.

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Review Exercise 28

1. Tick the correct option.

i. Inscribed angle of --------- is obtuse.
(a) minor arc (b) major arc (c) semi-circle (d) all of these

ii. ----------is the circular region bounded by arc and its chord
(a) Sector (b) Segment
(c) Major arc (d) Circumference

iii. If $2x$ is the measure of inscribed angle of minor arc then the central angle of corresponding major arc is ---------.
(a) x (b) $2x$ (c) $3x$ (d) $4x$

iv. If $2x$ and $60°$ are the measures of inscribed angles of same segment then x = -------.
(a) $120°$ (b) $60°$ (c) $20°$ (d) $30°$

v. The inscribed angle of minor arc of circle is ---------- angle.
(a) acute (b) obtuse
(c) right (d) reflex

vi. The inscribed angle of major arc of circle is ---------- angle.
(a) acute (b) obtuse
(c) right (d) reflex

vii. Sum of opposite angles of cyclic quadrilateral is ----------.
(a) $90°$ (b) $180°$ (c) $270°$ (d) $360°$

viii. A parallelogram inscribed in a circle is ----------.
(a) kite (b) trapezium
(c) rectangle (d) rhombus

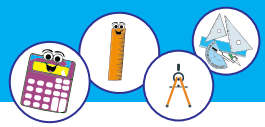
ix. The central angle of an arc is ----- than inscribed angle of corresponding arc.
(a) less (b) greater
(c) less or equal (d) greater or equal

x. The sum of central angles of all the arcs of a circle is ----------.
(a) $90°$ (b) $180°$ (c) $360°$ (d) $1000°$

SUMMARY

- The circular region bounded by an arc and two radial segments is called sector.
- The circular region bounded by an arc and its chord is called a segment.
- Angle in a segment of circle is also known as inscribed angle of the segment.
- Angle subtended by an arc at the centre of circle is called the central angle.
- The central angle of minor (or major) arc is double than inscribed angle of the corresponding major (or minor) arc respectively.
- Any two angles in the same segment of a circle are equal.
- The angle inscribed in semi-circle is right angle.
- The angle inscribed in major arc is acute.
- The angle inscribed in minor arc is obtuse.
- All the vertices of a cyclic quadrilateral are on the same circle.
- The opposite angles of cyclic quadrilateral are supplementary.

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2. Draw two unequal circles of radii 3.3cm and 2.1cm with centres, A and B respectively such that $m\overline{AB}$=8cm.
(a) Draw direct common tangents to these circles.
(b) Draw transverse common tangents to these circles.

3. Draw a tangent to two unequal circles of radii 3.8cm and 2.2cm with centres A and B respectively whereas
(i) circle touch internally
(ii) circles touch externally
(iii) circles intersect each other and $m\overline{AB}$=5.6cm.

4. Draw a circle which touches both the arms of an angle of measure
(i) 35° (ii) 40°

5. Draw a circle which passes through a point M which lies between two converging lines PQ and ST such that the circle also touches both the lines.

REVIEW EXERCISE 29

Tick the correct option

(i) Right bisectors of non-parallel_____ intersect at the centre of the circle.
(a) Radial segments (b) Chords (c) Tangents (d) none of these

(ii) _____circles can pass through a single point.
(a) One (b) Two (c) Three (d) Infinite

(iii)_____circles can pass through three non-collinear points.
(a) One (b) Two (c) Three (d) Infinite

(iv)_____angles of regular polygon are equal in measure.
(a) Interior (b) Exterior (c) both a and b (d) none of these

(v) Each interior angle of regular hexagon is equal to_____.
(a) 90° (b) 108° (c) 120° (d) 135°

(vi)A circle which touches all the sides of triangle is called_____.
(a) circumcircle (b) incircle (c) excircle (d) tricircle

(vii) A circle which touches one side of triangle externally and two produced sides internally is called_____.
(a) excircle (b) circumcircle (c) incircle (d) tricircle

(viii) The centre of inscribed circle is called_____
(a) excentre (b) incentre (c) centroid (d) orthocentre

(ix)Right bisectors of sides of a triangle intersect each other at_____
(a) incentre (b) excentre (c) centroid (d) circumcentre

(x) The point of intersection of internal bisectors of angles of a triangle is called
(a) excentre (b) circum centre (c) centroid (d) incentre

(xi)If a regular hexagon is inscribed in a circle then length of each side of hexagon _____ radius of the circle
(a) < (b) > (c) = (d) ≤

(xii) Angle between tangent and radial segment at the point of contact is _____ angle
(a) right (b) obtuse (c) acute (d) reflex

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(xiii) The central angle of circle joining the points of contact of tangents is _____ of the angle between the tangents

(a) complement (b) supplement (c) square (d) cube

(xiv) _____ common tangents do not intersect the line segment joining the centres of the circle

(a) internal (b) transverse (c) direct (d) vertical

(xv) Direct common tangents of two equal circles are _____

(a) intersecting (b) coincident (c) equal (d) parallel

(xvi) _____ common tangents of two equal circles intersect at the midpoint of the line segment joining the centres of the circles

(a) direct (b) transverse (c) external (d) parallel

(xvii) If two circles of radii 5cm and 2cm touch each other externally then the distance between their centers is _____

(a) 5cm (b) 10cm (c) 3cm (d) 7cm

(xviii) If two circles of radii 5cm and 2cm touch each other internally then the distance between their centres is_____

(a) 5cm (b) 10cm (c) 3cm (d) 7cm

(xix) Two or more converging lines always intersect each other at _____

(a) single point (b) two points

(c) more than one point (d) none of these

(xx) Three or more sided closed figure is called _____

(a) pentagon (b) hexagon (c) heptagon (d) polygon

SUMMARY

- Centre of a circle can be located by drawing the right bisectors of two non-parallel chords which meet each other at the centre.
- From three non-collinear points, a unique circle can be drawn.
- A circle can be completed without finding the centre when an arc is given with the help of regular polygon.
- Circumcircle always passes through all the vertices of a polygon.
- Incircle of a triangle always touches all the sides of the triangle.
- Excircle of a triangle touches one side of a triangle externally and two produced sides internally.
- Three excircles of a triangle can be constructed.
- Tangent is a line which touches circle at a single point
- Secant always cuts the circle at two points.
- Tangent to a circle is always perpendicular to the radial segment at the point of contact.
- Only one tangent can be drawn to a circle from a point of its circumference.
- Only two tangents to a circle can be drawn from a point outside the circle.
- The central angle of a circle joining the points of contact is supplement of the angle between the tangents.
- If a line is tangent to more than one circle, it is called common tangent.

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EXERCISE 30.5

1. From the top of a light house 102 meters high, measure of the angle of depression of a ship is $18°30'$. How far is the ship from the light house.
2. A ladder makes angle $60°$ with the ground and reaches a height of 6m on the wall, find the length of the ladder.
3. Find the angle of elevation when a 6m high bamboo makes a shadow of length $2\sqrt{3}$m.
4. An angle of elevation of the top of cliff is $30°$. Walking 210 m from the point towards the cliff, the angle of elevation is $45°$. Find the height of cliff.
5. An observation balloon is 4280m above the ground and 9613m away from a farm house. Find angle of depression of the farm house as observed from the balloon.

REVIEW EXERCISE 30

1. Multiple Choice Question M.C.Qs.
   Spot the correct option.

i. The system of measurement in which angle is measured in radian is called:

(a) CGS System (b) Sexagesimal
(c) MKS system (d) circular system

ii. The union of two non-collinear rays at common vertex is called

(a) an angle (b) a degree
(c) a minute (d) radian

iii. $10° =$ _____

(a) $600''$ (b) $3600''$ (c) $600'$ (d) 600

iv. $5\frac{\pi}{4}$ radians = _____

(a) $115°$ (b) $225°$ (c) $135°$ (d) $45°$

v. $\frac{1}{2}\sec 45° =$ _____

(a) $\sqrt{2}$ (b) $\frac{\sqrt{2}}{2}$ (c) $\frac{2}{\sqrt{2}}$ (d) $\frac{1}{2}$

vi. $\operatorname{cosec}\theta \cdot \sin\theta =$ _____

(a) 1 (b) 0 (c) -1 (d) 0.5

vii. $\sec^2\theta - \tan^2\theta =$ _____

(a) 0 (b) 1 (c) -1 (d) $\cos^2\theta$

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viii. In unit circle for point (x, y), $\sin\theta =$ _____

(a) x (b) $-x$ (c) y (d) $-y$

ix. If $\sin\theta = -\frac{1}{\sqrt{2}}$ and θ lies in the 3rd quadrant, then $\tan\theta$=?

(a) -1 (b) 1 (c) ± 1 (d) $\frac{1}{\sqrt{2}}$

x. If an object is above the level of an observer then angle formed between the horizontal line and observer's line of sight is called:

(a) an angle of depression (b) an angle of elevation

(c) an obtuse angle (d) None of these

2. Define an angle and angle in standard position.

3. Convert the $70°30'90''$ into radians.

4. How many degrees are there in $(7200)'$.

5. Covert $\frac{2\pi}{3}$ radians to degree measure.

6. Covert $\frac{3}{\pi}$ radians to D°M′S″ form.

7. Find r, when $l = 7\text{cm}$ and $\theta = \frac{\pi}{4}$ radian

8. Find θ, when arc length equal to radius of a circle.

9. Prove that $(1-\cos^2\theta)(1+\cot^2\theta) = 1$

10. Calculate area of the circular sector when $r = 2\text{cm}$ and $\theta = 3$ radians.

SUMMARY

- Division of circumference of a circle into 360 equal parts and angle subtended at centre by each part is called one degree, and is denoted by $1°$.
- The angle subtended at the centre of the circle such that arc length and radius of the circle are equal then angle measured is of one radian.
- Sub-division of radian/degree measures,
  - $1° = \frac{\pi}{180}$ radian ≈ 0.01745 radians and
  - $1° = \left(\frac{180°}{\pi}\right) \approx 57.3°$
- Formula for finding arc length of circle i.e. $l = r\theta$.
- Formula for area of sector $= \frac{1}{2}r^2\theta$ or $= \frac{1}{2}lr$.

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